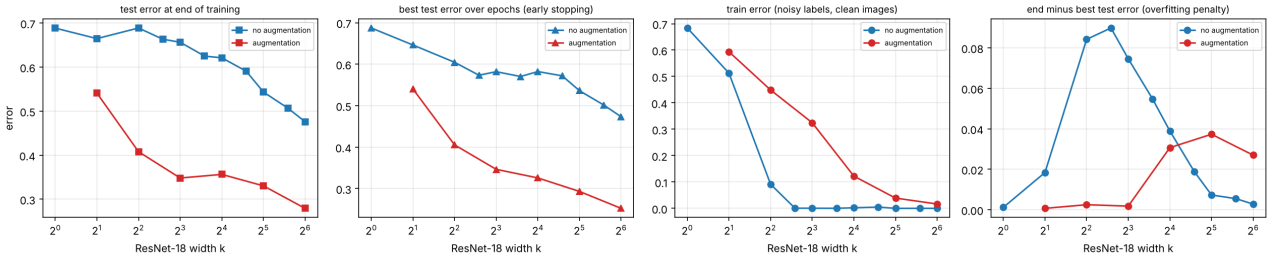
Tree-type networks revisited, part 2: augmentation, interpolation, ensembles, bias-variance and loss landscape

Ronak Raichandani (2026EEY7568). Code, results and notebooks:
https://github.com/ronakraichandani/MLOAssignment_2026EEY7568_Ronak

1. What was added (innovations)

(i) Augmentation as extra points to interpolate. The trunk is frozen, so each augmented view of an image is pushed through it once and the tree is grown on the union of the A views (view 0 = clean image). The exactness argument of the tree (every node interpolates its own masked problem, hence 100% training accuracy) is unchanged, so the tree over-fits the *augmented* set. For ResNet-18(k) we use random-crop + flip on the fly. **(ii) Test-time augmentation vote.** A tree answers by majority vote over T+1 views of the test image. **(iii) Majority-vote ensembles of interpolating trees** (ties broken by mean softmax mass, never overturning a strict majority) with three sources of diversity: bootstrap resampling, separately trained trunks and independent augmentation draws; a seed-only ablation separates them. **(iv) View-consistency label-noise detector:** a mislabelled image must be repaired in *every* view; the mean repair depth over views / fraction of views the root gets wrong is compared with the single-view repair depth. **(v) Bias-variance protocol** on disjoint training sets with the finite-M correction, comparing the interpolating tree, its own root node and one wide MLP, with and without augmentation, against n and against epochs. **(vi) Loss-landscape probes**: filter-normalised slices, sharpness, top Hessian eigenvalue and linear paths for ResNets; random weight perturbation of tree nodes (root only vs repair children only vs all).

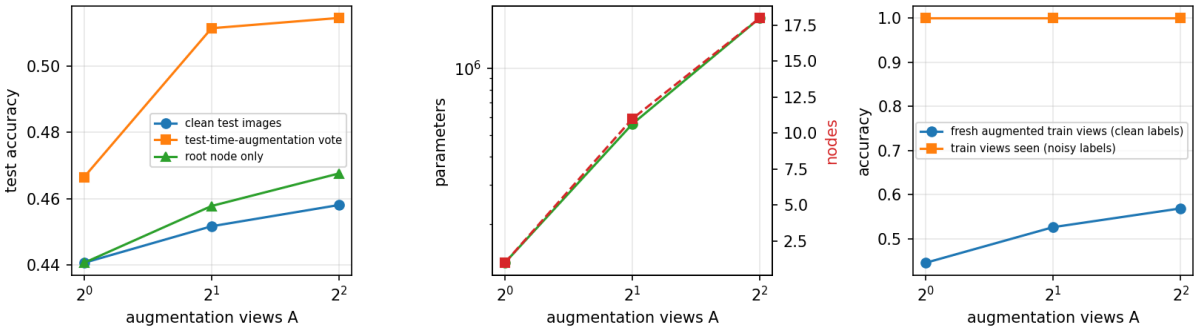
2. Augmentation + over-fitting: double descent and test accuracy



Width-k ResNet-18, 15k CIFAR-10 images, 15% symmetric label noise, Adam 1e-4, same seed and widths. Median of the last 3 evaluations for 'end of training'.

k	params	no aug: end	no aug: best	aug: end	aug: best	aug train err
2	11420	66.47%	64.63%	54.09%	54.02%	59.37%
4	44622	68.88%	60.46%	40.79%	40.54%	44.88%
8	176402	65.65%	58.19%	34.82%	34.64%	32.45%
16	701466	62.11%	58.22%	35.64%	32.57%	12.23%
32	2797610	54.34%	53.61%	33.04%	29.31%	3.89%
64	11173962	47.60%	47.32%	27.93%	25.23%	1.65%

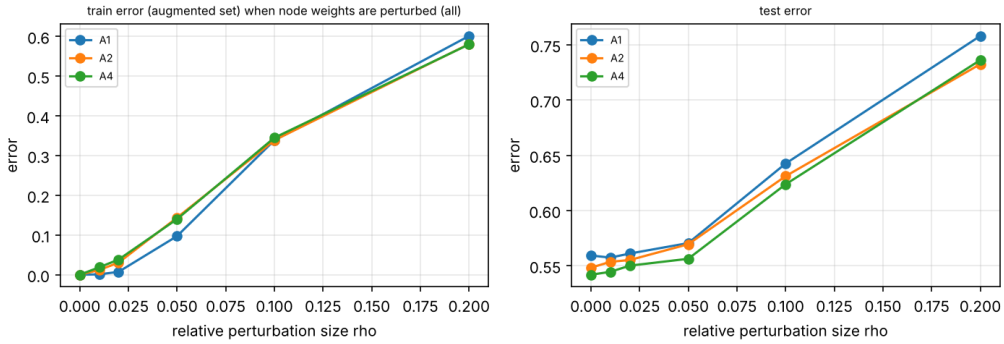
Test error (lower is better). 'best' = best epoch (needs a clean validation set, so it is an oracle early-stopping bound).



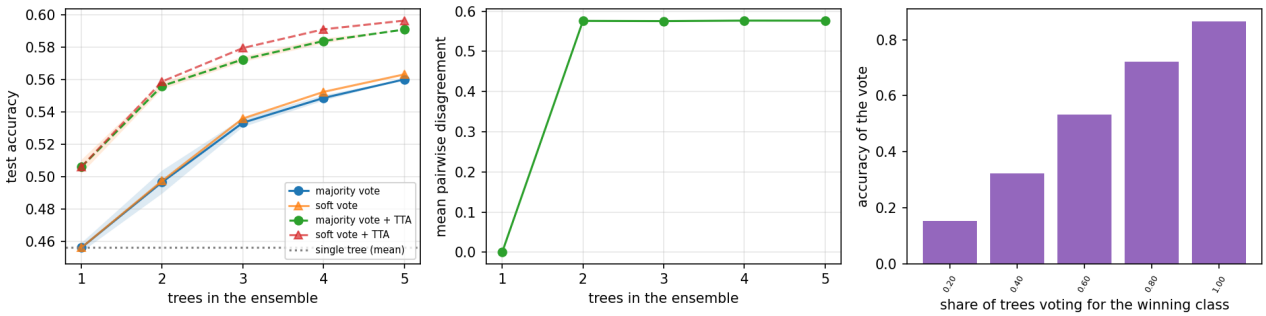
Single interpolating tree vs number of augmented views A.

A	nodes	params	train acc (all views)	test clean	test TTA vote	root only	fresh-aug train acc*	AUROC rep.depth	AUROC mean depth	AUROC root-miss
1	1	133898	100.00%	44.07%	46.65%	44.07%	44.62%	0.500	0.500	0.500
2	11	561576	100.00%	45.17%	51.13%	45.78%	52.68%	0.531	0.546	0.552
4	18	1680061	100.00%	45.81%	51.44%	46.76%	56.92%	0.583	0.653	0.679

*accuracy w.r.t. the clean labels on augmentations never used in training (did the tree learn invariance?). AUROC: label-noise detectors, 0.5 = chance.



3. Ensembles of trees with majority voting



M=5 interpolating trees. Left: ensemble accuracy vs size (mean over random subsets of members, band = 1 std). Middle: disagreement. Right: accuracy vs share of agreeing trees.

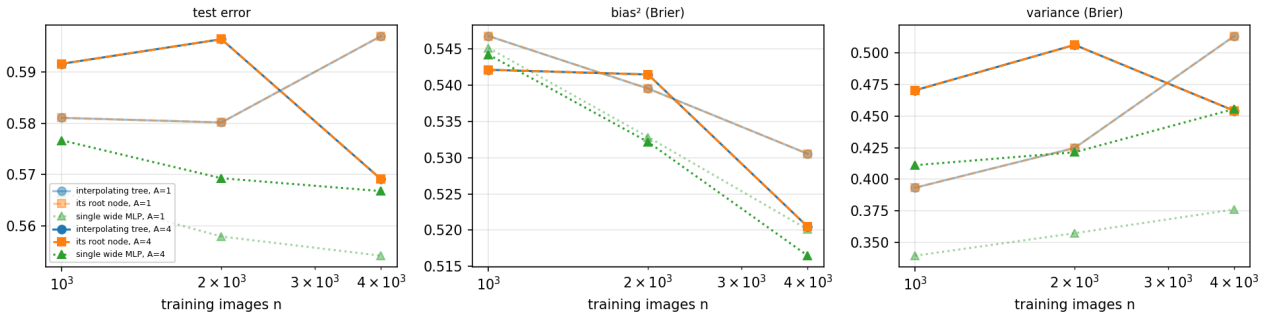
trees M	single (mean)	majority vote	soft vote	single + TTA	majority + TTA	pairwise disagreement
1	45.62%	45.62%	45.62%	50.61%	50.61%	0.000
2	45.65%	49.67%	49.75%	50.79%	55.61%	0.576
4	45.64%	54.86%	55.24%	50.76%	58.38%	0.577
5	45.63%	56.01%	56.32%	50.76%	59.09%	0.577

Oracle (some tree right): 80.66%. Mean parameters per tree 955986; ensemble total 4779930. 0-1 bias of the vote (its error) 0.452, variance of members around the vote 0.369.

Ablation, no augmentation (A=1), 5 trees: single 45.74% -> majority vote 56.41%; disagreement 0.570.

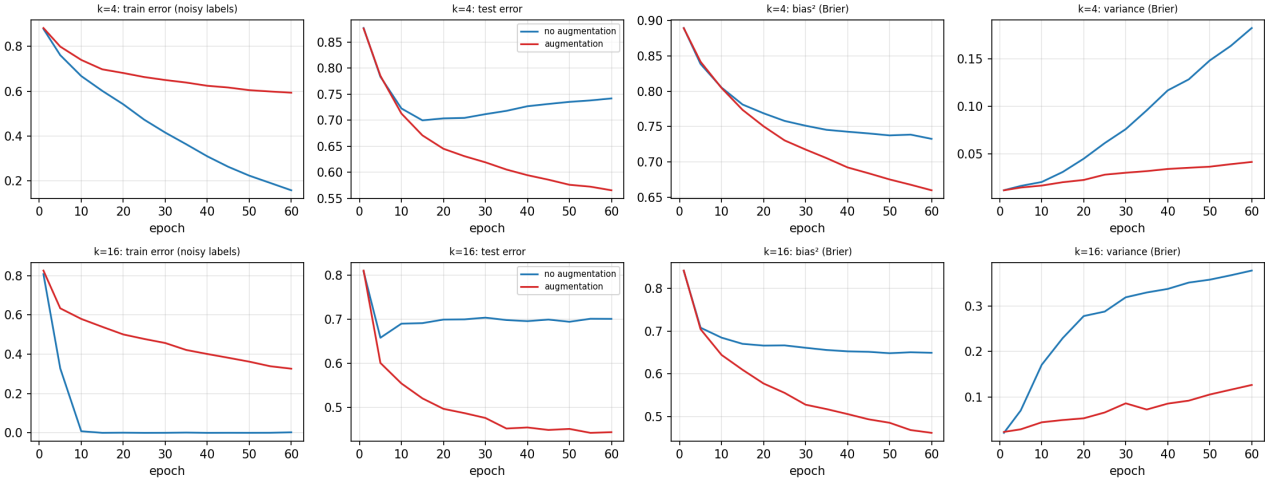
Ablation, diversity from seeds/augmentation draws only (same trunk, same data): single 45.74% -> majority vote of 5 trees 51.89%; disagreement 0.481.

4. Bias and variance vs number of samples and vs epochs



Brier decomposition on the clean test set; M independent training sets per n.

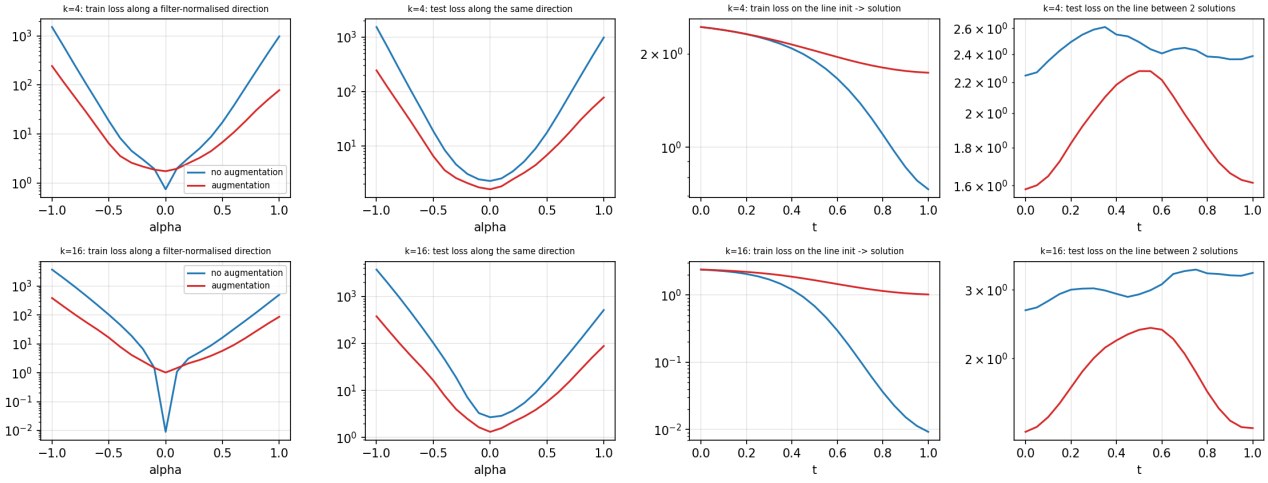
n	A	model	train acc	test err	bias^2	variance
1000	1	root node	100.00%	58.11%	0.547	0.393
1000	1	tree	100.00%	58.11%	0.547	0.393
1000	1	wide MLP	100.00%	56.65%	0.545	0.339
2000	1	root node	100.00%	58.02%	0.540	0.425
2000	1	tree	100.00%	58.02%	0.540	0.425
2000	1	wide MLP	100.00%	55.79%	0.533	0.357
4000	1	root node	100.00%	59.71%	0.531	0.513
4000	1	tree	100.00%	59.71%	0.531	0.513
4000	1	wide MLP	100.00%	55.41%	0.520	0.376
1000	4	root node	100.00%	59.16%	0.542	0.470
1000	4	tree	100.00%	59.16%	0.542	0.470
1000	4	wide MLP	100.00%	57.67%	0.544	0.411
2000	4	root node	100.00%	59.65%	0.541	0.506
2000	4	tree	100.00%	59.65%	0.541	0.506
2000	4	wide MLP	100.00%	56.93%	0.532	0.421
4000	4	root node	99.99%	56.91%	0.520	0.454
4000	4	tree	100.00%	56.91%	0.520	0.454
4000	4	wide MLP	100.00%	56.67%	0.516	0.456



ResNet-18(k) trained on n disjoint noisy subsets: bias and variance at every epoch.

config	params	M	best epoch: test err / bias^2 / var	end: test err / bias^2 / var
k16_aug0	701466	3	5: 0.658 / 0.708 / 0.071	0.701 / 0.649 / 0.378
k16_aug1	701466	3	55: 0.442 / 0.468 / 0.116	0.443 / 0.462 / 0.127
k4_aug0	44622	3	15: 0.700 / 0.781 / 0.031	0.742 / 0.733 / 0.182
k4_aug1	44622	3	60: 0.565 / 0.660 / 0.041	0.565 / 0.660 / 0.041

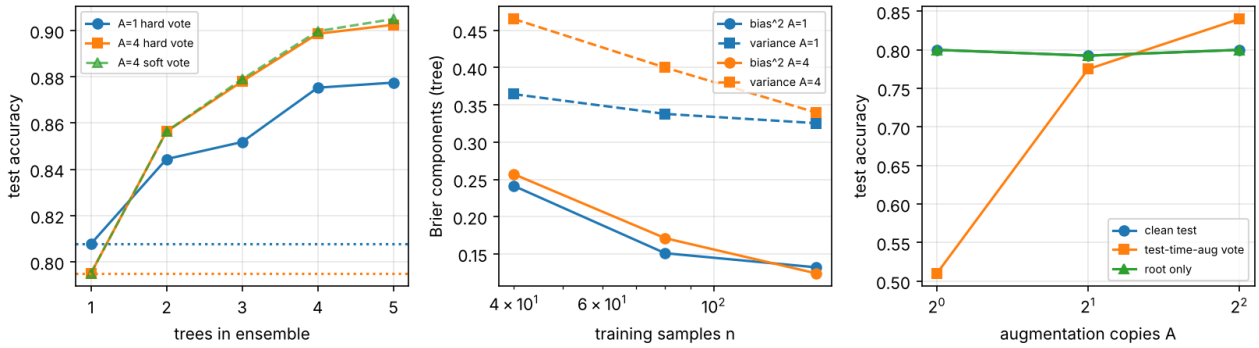
5. Loss landscape



Filter-normalised slices (log scale), line from initialisation to solution and line between two independently trained solutions (BatchNorm statistics re-estimated).

config	train loss	test err	top Hessian eig	d loss (rho=0.05)	test-loss barrier
k16_aug0	0.009	69.25%	920.2	0.153	0.392
k16_aug1	1.019	44.70%	3299.5	0.111	1.083
k4_aug0	0.745	72.65%	4352.6	0.490	0.293
k4_aug1	1.741	56.20%	1187.1	0.051	0.680

Appendix A. CPU pilot on 8x8 digits (same code paths, raw pixels + linear nodes)



Pilot only: tiny images, one split, 15% label noise. It verified the code paths before the Colab runs; it does not replace the CIFAR-10 results above.

6. Interpretations and conclusions

Augmentation and double descent

Setting: CIFAR-10, 15,000 training images, 15% noisy labels, ResNet-18(k), Adam 1e-4, 120 epochs, one seed. The baseline (no augmentation) is my earlier run with exactly the same recipe; the new run adds a random crop and flip drawn fresh every batch.

Without augmentation the usual picture appears. Test error (median of the last 3 evaluations) is 0.665 at k=2, jumps to 0.689 at k=4 (train error 0.09, the model is just reaching interpolation) and then falls to 0.476 at k=64. The epoch-wise penalty (end-of-training minus best-epoch test error) peaks at 0.090 at k=6 and shrinks to 0.003 for the widest net. This is both model-wise and epoch-wise double descent.

With augmentation the curve is lower everywhere and the peak is almost gone: 0.541, 0.408, 0.348 (k=8), 0.356 (k=16), 0.330, 0.279 (k=64). The only trace of a peak is a +0.008 bump at k=16, and the epoch-wise penalty grows from 0.002 (k=8) to 0.031 (k=16), 0.037 (k=32) and 0.027 (k=64). The training error stays above zero (0.12 at k=16, 0.017 at k=64), so the same widths that interpolated the plain data (k>=6) no longer interpolate the augmented stream: the interpolation threshold has moved to the right, to roughly k=16-32, and the penalty of memorising the noisy labels at that point (0.03-0.04) is about 2.5x smaller than without augmentation. The improvement is large: 0.279 versus 0.476 at k=64 (about 20 points) and 0.252 versus 0.473 for the best epoch. A clean second descent is therefore not reproduced; with one seed I cannot call the k=16 bump anything more than a hint (differences of 0.01 are within what I would expect from seed noise).

For the tree the over-fitting is made explicit: the frozen trunk produces A views per image and the tree is grown until it interpolates all A x 10,000 points, noisy labels included (every row has train accuracy 1.0000 and 100% of the noisy labels are memorised on the clean view). Test accuracy on clean images rises only slightly with A (0.441, 0.452, 0.458 for A=1, 2, 4), but averaging over 5 augmented test views (test-time augmentation, TTA) gives 0.467, 0.511, 0.514. Accuracy on fresh, never-seen augmentations of the training images (clean labels) rises 0.446, 0.527, 0.569: the tree learns invariance to the transformations, not only the exact views. Cost: nodes 1, 11, 18 and parameters 0.13M, 0.56M, 1.68M. The root alone is slightly better than the full tree on clean test images (0.468 versus 0.458 at A=4), the usual price of interpolating noise.

Ensembles with majority voting

Five trees (A=4), each with its own trunk and a bootstrap sample of the training images, each interpolating its augmented set, vote by hard majority (ties broken by mean softmax). A single tree gets 0.456; the majority vote of 5 gets 0.560 (soft vote 0.563), a gain of 10 points that is still rising at M=5 (0.497, 0.533, 0.549, 0.560 for M=2-5), so more trees would help further. Members disagree on 58% of test images yet are individually 0.45-0.46 accurate, and the best-of-5 oracle reaches 0.807, so the members' errors are largely different. The vote share is a calibrated confidence: images where all 5 trees agree are 86.6% correct, those with a 2-3 split 32-53%, and a 1-in-5 share only 15%.

With TTA each member answers with the mean of 5 augmented views and the majority vote reaches 0.591 (soft 0.596), versus 0.565 for the A=1 ensemble with TTA and 0.564 on clean images. So on clean test images augmentation adds nothing to the ensemble (0.560 versus 0.564); the gain from augmentation, about 2.6 points, only appears when it is also used at test time, and both effects add: single tree 0.456, + ensemble 0.560, + TTA 0.591. Where the diversity comes from matters: when the five trees share one trunk and differ only in node initialisation and augmentation draws (seed-only diversity), the members disagree on 48% of images instead of 58%, the oracle falls from 0.807 to 0.742, and the majority vote reaches only 0.519 (soft 0.518; with TTA 0.563 versus 0.591). Separate trunks and bootstrap resampling therefore give about 4 points more than augmentation noise and random initialisation alone, so the diversity of the features, not of the repair nodes, drives the ensemble gain. The A=4 ensemble has 4.8M parameters versus 0.67M for the A=1 ensemble, so the extra capacity buys little unless TTA is used. The ensembles are strongly dominated by variance reduction: the Domingos main prediction (the vote) has 0-1 bias 0.45 and the members have variance 0.37.

Bias and variance

Trees versus n (4 disjoint training sets per n, A=1 and A=4, Brier decomposition, finite-M corrected): the squared bias falls slowly with n (0.547, 0.540, 0.531 for A=1; 0.542, 0.541, 0.520 for A=4) while the variance does not

fall: 0.393, 0.425, 0.513 for A=1 and 0.470, 0.506, 0.454 for A=4. Interpolating models fit the noisy labels of whatever subset they see, so more data lowers the bias but does not shrink the variance. At A=4 and n=4000 the variance finally drops and the test error is the lowest (0.569 versus 0.597 for A=1). At these sizes the 64-wide root interpolates all A x n points by itself, so the tree equals its root (1.0-2.25 nodes on average); the repairing children appear only at A=4, n=4000. A wide one-hidden-layer MLP has the lowest error (0.554 to 0.567) and, for A=1, lower variance than the tree (0.376 versus 0.513 at n=4000).

ResNet-18(k) versus epochs (n=5000, 3 disjoint training sets, k=4 and 16, 60 epochs): without augmentation the variance grows steadily with training (k=16: 0.022, 0.071, 0.171, 0.378 at epochs 1, 5, 10, 60) while the bias keeps falling (0.842 to 0.649), and test error rises from its best 0.658 at epoch 5 to 0.701. This growth of variance when the training error reaches zero at epoch 10 is the epoch-wise over-fitting peak. With augmentation the variance reaches only 0.127 (k=16) and 0.041 (k=4) at epoch 60, the bias falls further (0.462) and test error keeps decreasing to 0.443 (k=16) and 0.565 (k=4), versus 0.701 and 0.742 without. The 3x lower variance is the mechanism by which augmentation prevents the peak. Only 3 draws were used, so individual numbers have noise of a few hundredths.

Loss landscape

For random filter-normalised directions the loss of the k=16 network without augmentation is 0.01 at the solution and rises to 3,771 and 508 at alpha=-1 and +1; the augmented net behaves the same way (389 and 87), so the solutions are sharp in absolute terms. The more informative numbers are the loss increases under a random perturbation of relative size 0.1: 1.05 (k=16) and 1.18 (k=4) without augmentation versus 0.40 and 0.15 with augmentation, so augmented solutions are 2.6-8x flatter on average over random directions. The largest Hessian eigenvalue does not agree (k=4: 4353 without versus 1187 with augmentation; k=16: 920 without versus 3300 with), so the improvement is in the typical directions, not the worst one. The augmented k=16 net also does not reach zero training loss (1.02), so the two are not strictly comparable. The linear path from the random initialisation to the solution decreases monotonically in all four runs, so there is no barrier on the way into the solution. The path between two independently trained networks (different data subsets) has test-loss barriers of 0.39 (k=16) and 0.29 (k=4) without augmentation and 1.08 and 0.68 with augmentation: the augmented solutions lie in basins that are less linearly connected, as expected for networks that must satisfy many more constraints.

For the interpolating tree, perturbing the weights by 10% (relative, filter-normalised) raises the training error on the augmented points to 0.34 (A=1) and 0.34 (A=4, all nodes), most of it from the root (0.29): the root carries the sharpness. Perturbing only the repair children raises the training error to 0.08 (A=4, rho=0.1, versus 0.29 for the root) and flips 11% of test predictions (root: 36%), so the children matter but less than the root, which carries most of the sharpness. The interpolating tree is therefore a sharp solution, with the root being the part that must stay accurate.

Conclusions and limitations

Augmentation lowered test error by about 20 points at the widest network, shifted and almost removed the double-descent peak, and cut the epoch-wise variance by about 3x; a tree grown on 4 augmented views and voting over 5 test views is 7 points better than a tree grown on clean images only (0.514 versus 0.441) and has learned invariance (accuracy on fresh views 0.446 to 0.569). Majority voting over 5 trees gives the largest single gain (+10 points) and TTA adds about 3 more, but augmentation alone does not help the clean-image ensemble. The loss landscape of augmented ResNets is flatter in random directions but not in the top Hessian direction. Limitations: one seed for the double-descent curves (a +0.008 bump is not significant); 3-5 independent draws for the bias-variance estimates; 10,000-image subsets and a frozen, weak trunk (about 46% training accuracy) for the tree experiments; the A=1 "no augmentation" ensemble still uses an augmentation-trained trunk; the k=4 and k=16 landscape runs are for non-interpolating augmented nets.